

Evidence Sufficiency Analysis: Quantifying the Semantic Gap Between Deep-Space Exploration Queries and Irrelevant Retrieval Results

Input Question

What is preventing humans from carrying out deep-space exploration?

Domain

Astronomy

Validation Status

needs_data

Problem Statement

The inquiry seeks to identify specific technological and physiological barriers hindering human deep-space exploration. However, the provided allowed evidence corpus consists exclusively of literature on DDoS attack prevention (EV-Q062-854247c1303697a41380bf80) and online discourse polarization (EV-Q062-4e6c7efb6fc4d59c2dfa012b). There is a critical mismatch between the question domain (Aerospace/Astronomy) and the available evidence domains (Cybersecurity/Social Science). Consequently, no factual claims regarding propulsion, life-support, or space physiology can be substantiated using the allowed evidence IDs.

Rationale

Strict adherence to SAGE125 principles requires that all factual claims trace back to allowed EvidenceCards. Since neither allowed evidence card contains information relevant to deep-space exploration, formulating a substantive scientific hypothesis about space travel barriers would require fabricating evidence or citing non-allowed sources (such as the booklet excerpt which lacks a valid evidence_id). Therefore, the only scientifically valid approach is to hypothesize that the current evidence set is insufficient and to design an experiment that validates this semantic gap.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Insufficient evidence prevents formulation of a valid scientific hypothesis regarding barriers to human deep-space exploration based on allowed sources.
- **Mechanism**: The allowed evidence IDs (EV-Q062-854247c1303697a41380bf80, EV-Q062-4e6c7efb6fc4d59c2dfa012b) pertain to DDoS prevention and online discourse polarization, respectively. No causal or correlational link exists between these domains and the astronomical/physiological

constraints of deep-space travel mentioned in the booklet excerpt. Therefore, no mechanism can be constructed from verified evidence.

- ****Falsifiable Prediction****: If a comprehensive review of the allowed evidence cards reveals zero mentions of propulsion, life-support, radiation, or psychological stressors relevant to spaceflight, then any hypothesis attributing specific barriers to these factors using these IDs is invalid.
- ****Required Observations****: Semantic analysis of full text of EV-Q062-854247c1303697a41380bf80 confirming absence of aerospace engineering content ; Semantic analysis of full text of EV-Q062-4e6c7efb6fc4d59c2dfa012b confirming absence of space physiology content
- ****Risk of Being Wrong****: Low risk for this null hypothesis; high risk for any alternative hypothesis attempting to force-fit irrelevant evidence IDs to the question domain.

Hypothesis 2

- ****Hypothesis****: Current literature retrieval systems fail to map deep-space exploration queries to relevant astronomical evidence, resulting in systematic knowledge gaps despite available domain-specific excerpts.
- ****Mechanism****: The mismatch between the question domain (Astronomy/Deep-Space) and retrieved evidence (Cybersecurity/Social Media) suggests a failure in the evidence extraction or indexing pipeline. This meta-hypothesis posits that the barrier is not physical but informational within the current AI scientist workflow.
- ****Falsifiable Prediction****: If re-running the evidence extraction with corrected domain filters yields at least one evidence card containing 'propulsion', 'life-support', or 'interstellar' with relevance > 0.8, this hypothesis is falsified.
- ****Required Observations****: Audit log of evidence retrieval query parameters for Q062 ; Re-execution of evidence search with strict astronomy domain constraints
- ****Risk of Being Wrong****: Moderate; the mismatch could be due to dataset limitations rather than retrieval failure.

Technical Details

This experiment is designed to validate the 'Insufficient Evidence' hypothesis. The core technical task is to perform a rigorous semantic gap analysis between the query domain (Deep-Space Exploration Barriers) and the provided evidence corpus (Cybersecurity/Social Science). The system will use vector space modeling and keyword extraction to quantify the topical distance. Specifically, we will embed the question text and the full text of allowed evidence cards (EV-Q062-854247c1303697a41380bf80, EV-Q062-4e6c7efb6fc4d59c2dfa012b) into a shared latent space using a pre-trained scientific language model (e.g., SciBERT or SPECTER). We will then calculate cosine similarity scores and check for the presence of domain-specific keywords (e.g., 'propulsion', 'radiation', 'life-support', 'delta-v') within the evidence texts. The hypothesis predicts that similarity scores will be below a relevance threshold (e.g., < 0.3) and keyword density will be zero.

Datasets

Source

```json

```
[
{
 "id": "EV-Q062-854247c1303697a41380bf80",
 "description": "Full text of arXiv:1208.3557 regarding DDoS prevention mechanisms.",
 "type": "text_corpus"
},
{
 "id": "EV-Q062-4e6c7efb6fc4d59c2dfa012b",
 "description": "Full text of arXiv:2606.18226 regarding online discourse polarization.",
 "type": "text_corpus"
}
]
'''
```

## Target

```
'''json
{
 "description": "Semantic relevance metrics and keyword presence/absence logs for the domain of 'Human Deep-Space Exploration'.",
 "type": "structured_metrics"
}
'''
```

## Paper Abstract

Background: Accurate scientific inquiry depends on the availability of domain-relevant evidence. This study addresses the question of barriers to human deep-space exploration. Methods: We analyzed the allowed evidence corpus (EV-Q062-854247c1303697a41380bf80, EV-Q062-4e6c7efb6fc4d59c2dfa012b) using semantic embedding models and keyword extraction techniques focused on aerospace engineering and space physiology. Validation Plan: We hypothesize that the provided evidence is topically irrelevant. We will validate this by measuring cosine similarity scores against a baseline of known aerospace literature and counting domain-specific keyword occurrences. Results: pending. The study aims to formally document the insufficiency of the current evidence set to answer the research question without fabrication.

## Methods

1. Text Extraction: Retrieve full text from provided URLs for both evidence IDs. 2. Embedding Generation: Use SPECTER model to embed question and evidence paragraphs. 3. Similarity Calculation: Compute max cosine similarity per document. 4. Keyword Analysis: Search for terms ['propulsion', 'power systems', 'life-support', 'physical stress', 'mental stress']. 5. Statistical Comparison: Compare results against random baseline distributions.

## Experiments

### Baselines

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```json
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[
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"Random Baseline: Cosine similarity scores between the question and randomly selected abstracts from unrelated fields (e.g., marine biology, ancient history).",

"Domain-Matched Baseline: Cosine similarity scores between the question and abstracts from known aerospace engineering papers (e.g., from NASA Technical Reports Server) to establish a positive control threshold."

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]
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### Metrics

```
```json
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[
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"Max Cosine Similarity Score: The highest semantic similarity found between the question and any segment of the allowed evidence.",

"Domain Keyword Density: The count of deep-space specific keywords per 1000 words in the evidence texts.",

"Relevance Classification Accuracy: Binary classification (Relevant/Irrelevant) performance if a classifier were trained on these features, evaluated via cross-validation on a labeled dataset of relevant vs. irrelevant papers."

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]
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### Ablation

Remove the 'Social Science' evidence (EV-Q062-4e6c7efb6fc4d59c2dfa012b) and re-evaluate similarity to determine if the 'Cybersecurity' evidence (EV-Q062-854247c1303697a41380bf80) alone contributes any spurious semantic overlap.

### Validation Protocol

Double-blind review of the top 5 most similar paragraphs identified by the algorithm to ensure that high similarity scores are not due to generic scientific terminology (e.g., 'system', 'model', 'analysis') but rather domain-specific content. If generic terms drive similarity, adjust embedding weights to penalize common stop-words and generic

academic phrases.

## Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

## References

- **\*\*EV-Q062-854247c1303697a41380bf80\*\*** · arxiv · arXiv:1208.3557
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1208.3557.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=2847650785e6aeff3693aef33e18cc4b5930f39501182e534a67b2cbf8079ac5
- **\*\*EV-Q062-4e6c7efb6fc4d59c2dfa012b\*\*** · arxiv · arXiv:2606.18226
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2606.18226.pdf> · doi: Not available
- reliability\_note: eligibility\_status=FULLTEXT\_VERIFIED;  
topic\_relevance\_status=DIRECT\_QUESTION\_CORE; locator=page:1|section:page-1|paragraph:1;  
content\_sha256=8789f70f03ae7518ed5de4e76e7bcf0f24155e2cdc2011dde810d20bc4c29c5a

## Reviewer Comments

- The candidate hypothesis correctly identifies 'insufficient\_evidence' given that the allowed evidence IDs (DDoS prevention and online polarization) are topically irrelevant to deep-space exploration barriers.
- The system strictly adhered to Rule 10 by refusing to cite the booklet excerpt as factual evidence since it lacks a valid evidence\_id in the allowed list.
- The experiment design appropriately pivots to a meta-validation task (semantic gap analysis) rather than fabricating scientific results about space travel.
- Results field is correctly marked as pending/unexecuted, avoiding fabrication.
- No causal links were invented between cybersecurity/social media concepts and aerospace engineering.

## Revision History

## Reproducibility Checklist

- Verify access to full-text PDFs for EV-Q062-854247c1303697a41380bf80 and EV-Q062-4e6c7efb6fc4d59c2dfa012b.

- Confirm version of the sentence-transformer model used for embeddings.
- Archive the compiled keyword dictionary for deep-space exploration terms.
- Save raw cosine similarity matrices and keyword match logs.
- Document the random seed used for any stochastic processes in embedding or baseline generation.